

CPSC-406 Report

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Abstract

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1 Introduction

2 Week by Week

2.1 Week 1

Exercise 1

Consider the following DFAs $\mathcal{A}_1$ and $\mathcal{A}_2$.

Complete the following table with all 8 possible strings of length 3 over $\{a, b\}$:

String	Accepted by $\mathcal{A}_1$?	Accepted by $\mathcal{A}_2$?
aaa	No	Yes
aab	Yes	Yes
aba	Yes	No
abb	Yes	No
baa	No	Yes
bab	No	No
bba	No	No
bbb	No	No

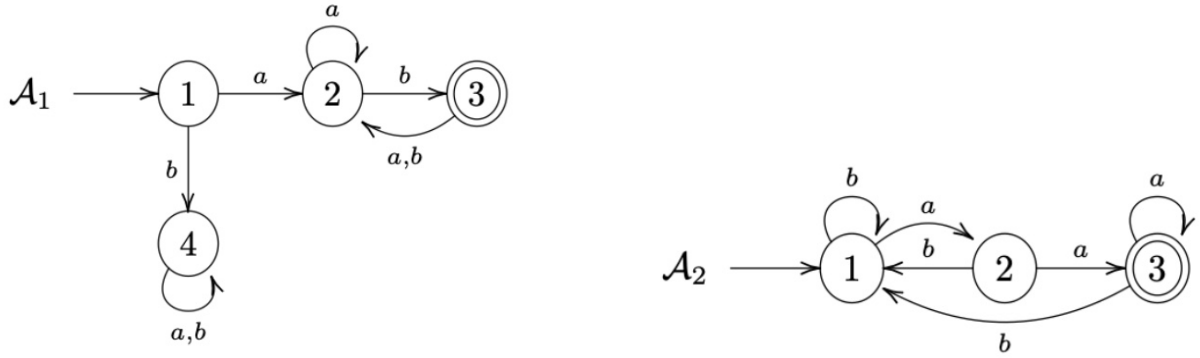


Figure 1: DFAs $\mathcal{A}_1$ (left) and $\mathcal{A}_2$ (right).

Exercise 1.2

Completely describe the languages $L(\mathcal{A}_k)$ accepted by $\mathcal{A}_k$ for $k = 1, 2$.

Answer:

$L(\mathcal{A}_1)$: The language accepted by $\mathcal{A}_1$ consists of all strings over $\{a, b\}$ that start with a and contain at least one b . In other words,

$$L(\mathcal{A}_1) = \{w \in \{a, b\}^* \mid w \text{ starts with } a \text{ and } w \text{ contains at least one } b\}.$$

Equivalently, $L(\mathcal{A}_1) = a(a \cup b)^*b(a \cup b)^*$: the first symbol must be a , and at least one b must appear somewhere in the string. (Any string beginning with b leads to the trap state 4 and is rejected.)

$L(\mathcal{A}_2)$: The language accepted by $\mathcal{A}_2$ consists of all strings over $\{a, b\}$ that contain the substring aa . That is,

$$L(\mathcal{A}_2) = \{w \in \{a, b\}^* \mid w \text{ contains the substring } aa\}.$$

Equivalently, $L(\mathcal{A}_2) = (a \cup b)^*aa(a \cup b)^*$: the automaton reaches the accepting state 3 when it reads two consecutive a 's.

3 Synthesis

4 Evidence of Participation

5 Conclusion

References

[BLA] Author, [Title](#), Publisher, Year.